

MATH-1014 T10B

Midterm Exercise(1)

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Definite Integrals (1)

Problem 1

Evaluate the integral

$$\int_0^{\frac{\pi}{2}} 6 \cos^3 x \sin^3 x \, dx.$$

Problem 2

Evaluate the integral

$$\int_1^2 \ln \frac{x+1}{x} \, dx.$$

Solution to Problem 1

Evaluate $I = \int_0^{\frac{\pi}{2}} 6 \cos^3 x \sin^3 x \, dx$.

1. Substitution:

Split off one $\cos x$ and use $\cos^2 x = 1 - \sin^2 x$. Let $u = \sin x$, then $du = \cos x \, dx$.

Bounds: $x = 0 \rightarrow u = 0$; $x = \frac{\pi}{2} \rightarrow u = 1$.

$$I = \int_0^{\frac{\pi}{2}} 6 \sin^3 x (1 - \sin^2 x) \cos x \, dx$$

2. Evaluate the new integral:

$$\begin{aligned} I &= \int_0^1 6u^3(1 - u^2) \, du \\ &= 6 \int_0^1 (u^3 - u^5) \, du \\ &= 6 \left[\frac{u^4}{4} - \frac{u^6}{6} \right]_0^1 \\ &= 6 \left(\frac{1}{4} - \frac{1}{6} \right) = \boxed{\frac{1}{2}} \end{aligned}$$

Solution to Problem 2

Evaluate $I = \int_1^2 \ln\left(\frac{x+1}{x}\right) dx$.

Integration by Parts:

Let $u = \ln(x+1) - \ln x \implies du = \left(\frac{1}{x+1} - \frac{1}{x}\right) dx = \frac{-1}{x(x+1)} dx$.

Let $dv = dx \implies v = x$.

$$\begin{aligned}
 I &= \left[x \ln \frac{x+1}{x} \right]_1^2 - \int_1^2 x \left(\frac{-1}{x(x+1)} \right) dx \\
 &= \left(2 \ln \frac{3}{2} - \ln 2 \right) + \int_1^2 \frac{1}{x+1} dx \\
 &= 2 \ln 3 - 3 \ln 2 + \left[\ln(x+1) \right]_1^2 \\
 &= 2 \ln 3 - 3 \ln 2 + (\ln 3 - \ln 2) \\
 &= 3 \ln 3 - 4 \ln 2 = \boxed{\ln \frac{27}{16}}
 \end{aligned}$$

Definite Integrals (2)

Problem 3

Evaluate the integral

$$\int_{\ln 3}^{\ln 7} \frac{5e^x}{(e^x - 2)(e^x + 3)} dx.$$

Problem 4

Evaluate the integral

$$\int_0^2 (x^4 - 1)\sqrt{4 - x^2} dx.$$

Solution to Problem 3

Evaluate $I = \int_{\ln 3}^{\ln 7} \frac{5e^x}{(e^x-2)(e^x+3)} dx$.

1. Substitution:

Let $u = e^x \implies du = e^x dx$. Bounds: $x = \ln 3 \rightarrow u = 3$; $x = \ln 7 \rightarrow u = 7$.

$$I = \int_3^7 \frac{5}{(u-2)(u+3)} du$$

2. Partial Fractions:

$$\frac{5}{(u-2)(u+3)} = \frac{1}{u-2} - \frac{1}{u+3}$$

3. Evaluate:

$$\begin{aligned} I &= \int_3^7 \left(\frac{1}{u-2} - \frac{1}{u+3} \right) du = \left[\ln |u-2| - \ln |u+3| \right]_3^7 \\ &= (\ln 5 - \ln 10) - (\ln 1 - \ln 6) \\ &= -\ln 2 + \ln 6 = \ln \left(\frac{6}{2} \right) = \boxed{\ln 3} \end{aligned}$$

Solution to Problem 4 (Part 1)

Evaluate $I = \int_0^2 (x^4 - 1)\sqrt{4 - x^2} dx$.

1. Trigonometric Substitution:

Let $x = 2 \sin \theta \implies dx = 2 \cos \theta d\theta$. Bounds: $0 \rightarrow \frac{\pi}{2}$.

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} (16 \sin^4 \theta - 1) \sqrt{4 - 4 \sin^2 \theta} \cdot 2 \cos \theta d\theta \\ &= \int_0^{\frac{\pi}{2}} (16 \sin^4 \theta - 1)(4 \cos^2 \theta) d\theta \\ &= 64 \int_0^{\frac{\pi}{2}} \sin^4 \theta \cos^2 \theta d\theta - 4 \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta \end{aligned}$$

2. Evaluate the easier integral: Using $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$:

$$\int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta = \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\frac{\pi}{2}} = \frac{\pi}{4}$$

Solution to Problem 4 (Part 2)

3. Evaluate the remaining integral:

Apply half-angle formulas to $\sin^4 \theta \cos^2 \theta = \sin^2 \theta (\sin \theta \cos \theta)^2$:

$$\begin{aligned}\sin^4 \theta \cos^2 \theta &= \left(\frac{1 - \cos 2\theta}{2} \right) \left(\frac{\sin 2\theta}{2} \right)^2 \\ &= \frac{1}{8} (\sin^2 2\theta - \sin^2 2\theta \cos 2\theta)\end{aligned}$$

Integrate term by term over $[0, \frac{\pi}{2}]$:

- $\int_0^{\frac{\pi}{2}} \sin^2 2\theta \, d\theta = \int_0^{\frac{\pi}{2}} \frac{1 - \cos 4\theta}{2} \, d\theta = \frac{\pi}{4}$
- $\int_0^{\frac{\pi}{2}} \sin^2 2\theta \cos 2\theta \, d\theta = \left[\frac{\sin^3 2\theta}{6} \right]_0^{\frac{\pi}{2}} = 0$

Thus, $\int_0^{\frac{\pi}{2}} \sin^4 \theta \cos^2 \theta \, d\theta = \frac{1}{8} \left(\frac{\pi}{4} - 0 \right) = \frac{\pi}{32}$.

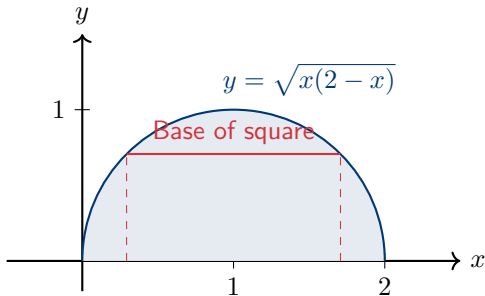
4. Final Result:

$$I = 64 \left(\frac{\pi}{32} \right) - 4 \left(\frac{\pi}{4} \right) = 2\pi - \pi = \boxed{\pi}$$

Volumes by Cross Sections

Problem 5

The base of a solid sitting on the xy -plane is the region enclosed by the graph of $y = \sqrt{x(2-x)}$ and the x -axis. Suppose that each cross section of the solid perpendicular to the y -axis is a square. Find the volume of the solid.



Solution to Problem 5

1. Base Region & Side Length:

Squaring $y = \sqrt{x(2-x)}$ gives $y^2 = 2x - x^2 \implies (x-1)^2 + y^2 = 1$.

This is a semicircle of radius 1 centered at $(1, 0)$, for $y \in [0, 1]$.

For a cross section perpendicular to the y -axis, the horizontal width is s . From the circle equation:

$$(x-1)^2 = 1 - y^2 \implies x = 1 \pm \sqrt{1 - y^2}$$

The right boundary is $x_R = 1 + \sqrt{1 - y^2}$, left is $x_L = 1 - \sqrt{1 - y^2}$.

Square side length: $s = x_R - x_L = 2\sqrt{1 - y^2}$.

2. Calculate the Volume:

Area of the square is $A(y) = s^2 = 4(1 - y^2)$.

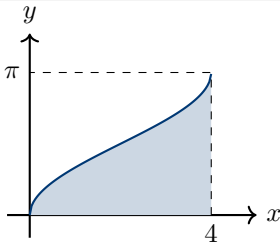
$$V = \int_0^1 A(y) dy = \int_0^1 4(1 - y^2) dy = 4 \left[y - \frac{y^3}{3} \right]_0^1 = \boxed{\frac{8}{3}}$$

Solid of Revolution and Work

Problem 6

The area under the graph of $y = \cos^{-1}\left(1 - \frac{x}{2}\right)$, where $0 \leq x \leq 4$ (meters), is rotated about the y -axis.

- (a) Express the **volume** of the solid as an integral. (**Do not evaluate**)
- (b) Express the **surface area** rotated about the y -axis as an integral.
- (c) If holding water up to 2 meters deep, express the **work** to pump all water to the top of the tank as an integral.



Solution to Problem 6 (a) & (b)

First, express the curve in terms of y :

$$y = \cos^{-1} \left(1 - \frac{x}{2} \right) \implies \cos y = 1 - \frac{x}{2} \implies x = 2 - 2 \cos y$$

Bounds: as $x \in [0, 4]$, $y \in [0, \pi]$.

(a) Volume (Disk Method about y -axis):

Radius $R(y) = x = 2 - 2 \cos y$.

$$V = \int_0^{\pi} \pi (2 - 2 \cos y)^2 dy$$

(b) Surface Area (Rotation about y -axis):

$x' = 2 \sin y$. Formula: $S = \int 2\pi x \sqrt{1 + (x')^2} dy$.

$$S = \int_0^{\pi} 2\pi (2 - 2 \cos y) \sqrt{1 + 4 \sin^2 y} dy$$

Solution to Problem 6 (c)

(c) Work to pump the water:

- **Water level:** $y = 0$ to $y = 2$.
- **Tank top:** $y = \pi$ meters.
- **Slice volume:** $dV = \pi x^2 dy = \pi(2 - 2 \cos y)^2 dy$.
- **Slice weight (Force):** $dF = \rho g dV$.
- **Lift distance:** Pumped to the top $\implies (\pi - y)$.

Integrate (Force $\times$ Distance) over the water bounds:

$$W = \int dF \cdot \text{distance}$$

$$W = \int_0^2 \rho g (\pi - y) \pi (2 - 2 \cos y)^2 dy$$

Reduction Formulas

Problem 7

Let $I_n = \int_{-1}^1 (1 - x^2)^n \cos(\pi x) dx$, where $n \geq 0$ is an integer.

- (a) Evaluate I_1 .
- (b) Apply integration by parts to show that, for every integer $n \geq 2$,

$$\pi^2 I_n = 2n(2n - 1)I_{n-1} - 4n(n - 1)I_{n-2}$$

- (c) Evaluate I_2 .

Solution to Problem 7 (a)

(a) Evaluate I_1

$$I_1 = \int_{-1}^1 (1 - x^2) \cos(\pi x) dx$$

Apply Integration by Parts: Let $u = 1 - x^2 \implies du = -2x dx$, and $dv = \cos(\pi x) dx \implies v = \frac{1}{\pi} \sin(\pi x)$.

$$\begin{aligned} I_1 &= \left[(1 - x^2) \frac{\sin(\pi x)}{\pi} \right]_{-1}^1 - \int_{-1}^1 \frac{\sin(\pi x)}{\pi} (-2x) dx \\ &= 0 + \frac{2}{\pi} \int_{-1}^1 x \sin(\pi x) dx \end{aligned}$$

Apply IBP again: Let $u = x \implies du = dx$, and $dv = \sin(\pi x) dx \implies v = -\frac{1}{\pi} \cos(\pi x)$.

$$\begin{aligned} I_1 &= \frac{2}{\pi} \left(\left[-x \frac{\cos(\pi x)}{\pi} \right]_{-1}^1 + \frac{1}{\pi} \int_{-1}^1 \cos(\pi x) dx \right) \\ &= \frac{2}{\pi} \left(\left(\frac{1}{\pi} - \left(-\frac{1}{\pi} \right) \right) + \frac{1}{\pi} \left[\frac{\sin(\pi x)}{\pi} \right]_{-1}^1 \right) = \frac{2}{\pi} \left(\frac{2}{\pi} + 0 \right) = \boxed{\frac{4}{\pi^2}} \end{aligned}$$

Solution to Problem 7 (b) - Part 1

(b) Show $\pi^2 I_n = 2n(2n-1)I_{n-1} - 4n(n-1)I_{n-2}$

$$I_n = \int_{-1}^1 (1-x^2)^n \cos(\pi x) dx$$

Apply IBP: $u = (1-x^2)^n \implies du = n(1-x^2)^{n-1}(-2x) dx$, and
 $dv = \cos(\pi x) dx \implies v = \frac{1}{\pi} \sin(\pi x)$.

$$\begin{aligned} I_n &= \left[(1-x^2)^n \frac{\sin(\pi x)}{\pi} \right]_{-1}^1 + \frac{2n}{\pi} \int_{-1}^1 x(1-x^2)^{n-1} \sin(\pi x) dx \\ &= 0 + \frac{2n}{\pi} \int_{-1}^1 x(1-x^2)^{n-1} \sin(\pi x) dx \end{aligned}$$

Apply IBP again to this new integral. Let $dv = \sin(\pi x) dx \implies v = -\frac{1}{\pi} \cos(\pi x)$.
 Let $u = x(1-x^2)^{n-1}$. By product and chain rules:

$$\begin{aligned} du &= [1 \cdot (1-x^2)^{n-1} + x \cdot (n-1)(1-x^2)^{n-2}(-2x)] dx \\ &= [(1-x^2)^{n-1} - 2x^2(n-1)(1-x^2)^{n-2}] dx \end{aligned}$$

Solution to Problem 7 (b) - Part 2

Substitute $2x^2 = 2 - 2(1 - x^2)$ into du :

$$\begin{aligned} du &= [(1 - x^2)^{n-1} - (2 - 2(1 - x^2))(n - 1)(1 - x^2)^{n-2}] dx \\ &= [(1 - x^2)^{n-1} - 2(n - 1)(1 - x^2)^{n-2} + 2(n - 1)(1 - x^2)^{n-1}] dx \\ &= [(2n - 1)(1 - x^2)^{n-1} - 2(n - 1)(1 - x^2)^{n-2}] dx \end{aligned}$$

Now finish the IBP: $\int u dv = [uv] - \int v du$. For $n \geq 2$, the boundary term

$$\left[-x(1 - x^2)^{n-1} \frac{\cos(\pi x)}{\pi} \right]_{-1}^1 = 0.$$

$$I_n = \frac{2n}{\pi} \left(0 + \frac{1}{\pi} \int_{-1}^1 \cos(\pi x) [(2n - 1)(1 - x^2)^{n-1} - 2(n - 1)(1 - x^2)^{n-2}] dx \right)$$

$$\pi^2 I_n = 2n(2n - 1) \int_{-1}^1 (1 - x^2)^{n-1} \cos(\pi x) dx$$

$$- 4n(n - 1) \int_{-1}^1 (1 - x^2)^{n-2} \cos(\pi x) dx$$

$$\pi^2 I_n = \boxed{2n(2n - 1)I_{n-1} - 4n(n - 1)I_{n-2}}$$

Solution to Problem 7 (c)

(c) Evaluate I_2

From part (b), we use the recurrence relation with $n = 2$:

$$\pi^2 I_2 = 2(2)(3)I_1 - 4(2)(1)I_0 = 12I_1 - 8I_0$$

We know $I_1 = \frac{4}{\pi^2}$. We need to calculate I_0 :

$$I_0 = \int_{-1}^1 (1 - x^2)^0 \cos(\pi x) dx = \int_{-1}^1 \cos(\pi x) dx$$

$$I_0 = \left[\frac{\sin(\pi x)}{\pi} \right]_{-1}^1 = 0$$

Substitute I_1 and I_0 into the relation:

$$\pi^2 I_2 = 12 \left(\frac{4}{\pi^2} \right) - 8(0)$$

$$\pi^2 I_2 = \frac{48}{\pi^2}$$

$$I_2 = \boxed{\frac{48}{\pi^4}}$$

Strategy: Integrals of $\sin^m x \cos^n x$

Consider the integral $\int \sin^m x \cos^n x dx$.

- **Case 1: m is odd** ($m = 2k + 1$)

Split off one $\sin x$ and convert the rest to cosines:

$$\sin^m x = (\sin^2 x)^k \sin x = (1 - \cos^2 x)^k \sin x$$

Then use the substitution $u = \cos x$.

- **Case 2: n is odd** ($n = 2k + 1$)

Split off one $\cos x$ and convert the rest to sines:

$$\cos^n x = (\cos^2 x)^k \cos x = (1 - \sin^2 x)^k \cos x$$

Then use the substitution $u = \sin x$.

- **Case 3: Both m and n are even**

Use half-angle identities repeatedly to reduce the powers:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

Pro Tip: Try to bundle terms first if possible, e.g., $\sin^2 x \cos^2 x = \frac{1}{4} \sin^2 2x$

Strategy: Integrals of $\tan^m x \sec^n x$

Consider the integral $\int \tan^m x \sec^n x dx$.

- Case 1: n (power of secant) is even**

Split off $\sec^2 x$ and convert the remaining secants to tangents:

$$\sec^n x = (\sec^2 x)^k \sec^2 x = (1 + \tan^2 x)^k \sec^2 x$$

Then use the substitution $u = \tan x$ (since $du = \sec^2 x dx$).

- Case 2: m (power of tangent) is odd**

Split off $\sec x \tan x$ and convert the remaining tangents to secants:

$$\tan^m x = (\tan^2 x)^k \tan x = (\sec^2 x - 1)^k \tan x$$

Then use the substitution $u = \sec x$ (since $du = \sec x \tan x dx$).

- Case 3: m is even and n is odd**

Convert all tangents to secants using $\tan^2 x = \sec^2 x - 1$. You will be left with odd powers of $\sec x$, which usually require **Integration by Parts** (e.g., the classic $\int \sec^3 x dx$).

Strategy: Trigonometric Substitution

For integrals involving specific radical forms, use the following substitutions to simplify the integrand into pure trigonometric functions:

Radical Form	Substitution	Identity Used
$\sqrt{a^2 - x^2}$	$x = a \sin \theta, \quad dx = a \cos \theta \, d\theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta, \quad dx = a \sec^2 \theta \, d\theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta, \quad dx = a \sec \theta \tan \theta \, d\theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

Important Note

After integrating in terms of θ , remember to draw a **right triangle** based on your initial substitution to convert the final answer back to terms of x .